

Secure LoRa-Based Communication Using STM32F042

Course Information

- **Course:** ENGS 290-2 — Special Topics: Applied Cryptography for Embedded Systems
- **Semester:** Fall 2026

Group Members

- Anna Grigoryan — GR7204664
- Mher Saribekyan — SA0012566
- Avet Tonoyan — TO1261500
- Lusine Khachatryan — KH5540632

Proposed Project Title

Secure LoRa-Based Communication Using STM32F042

Project Description

This project aims to implement a secure wireless communication system using STM32F042 microcontroller and LoRa modules. Data transmitted between the devices will be protected using cryptographic techniques to ensure the confidentiality and integrity of the communication. The receiver will decrypt and verify the received data to ensure that messages have not been modified or tampered with during transmission. The project will demonstrate the practical application of cryptographic techniques in an embedded system communicating over a LoRa radio link.